

Module 1

Network - group of connected devices

use IP and MAC addresses to make sure communication happens with the right devices

LAN - Local area network

WAN - Wide area network

Network tools

Hub - broadcasts to every device

Switch - makes connections between specific devices
↳ control traffic flow

Router - connects multiple networks

Modem - router to internet

These are physical devices but their features can often be virtualized

Cloud networks

Nowadays more companies use cloud services for networking.

Cloud network - collection of servers or computers that stores resources and data in remote centers accessed via the internet

SaaS - applications

IaaS - physical resources, servers, storage

PaaS - operating systems, custom environments

Communication

Packet - unit of information, sent between devices.

Packets contain header, body, footer

Bandwidth - the amount of data a device receives every second

Sniffing - capturing and inspecting network packets

The TCP/IP model

TCP (Transmission Control Protocol)

IP (Internet Protocol)

Port - software based location that organizes sending and receiving data

4. Application → How packets interact
3. Transport → Protocols and status
2. Internet → IP addresses assigned, network connections
1. Link → Physical connections

The OSI model

Application - top level protocol (http, smtp, etc)

Presentation - formatting, encryption

Session - authentication, reconnection

Transport - speed, flow

Network - receive and deliver

Link - sending and receiving

Physical - Physical wiring

IP addresses are unique and identify devices on the internet

IPv4 addresses are x.x.x.x decimal

IPv6 addresses are xxxx:xxxx ... hex

They can be public or private. All on a network share the same public IP. A mac address is assigned to each local device and used for internet communication

IPv4 Packets

- max size 65,535 bytes
- the header contains the following info:
 - ↳ version VER
 - ↳ IP header length HLEN / IHL
 - ↳ Type of service Tos
 - ↳ Total length
 - ↳ Identification
 - ↳ Flags
 - ↳ Fragmentation offset
 - ↳ Time to live TTL
 - ↳ Protocol
 - ↳ Header checksum
 - ↳ Source IP
 - ↳ Dest IP
 - ↳ Options

IPv6 packets are considerably simpler.

Module 2

Protocols

Sets of rules that describe how data should be handled when sending/receiving.

Address resolution protocol (ARP) - used to determine the MAC address of the next router or device

Communication protocols

Management protocols

Security protocols

IEEE 802.11

WIFI protocols

WPA Security protocol for connecting over wifi

WEP → WPA → WPA2 → WPA3

↙ ↘
enterprise personal

Firewalls and Proxies

Network security device that monitors traffic and can block or allow it through the network.

Port Filtering - block ports but are uncommon to prevent unwanted communication

There are software and hardware firewalls. Software ones are generally cheaper. Companies may use cloud firewalls

Statefull - firewalls analyse and block

Stateless - uses rules to accept or block

↳ considered less secure

NGFW - next-gen firewall even more secure than stateful

↳ can use deep packet inspection.

VPN - changes public IP address by encapsulating traffic

Zone

Segment of a network, protecting the internal network

networks can be divided into subnets

Uncontrolled zone - outside networks, not controlled by org

Controlled zone - owned by organisation

↳ DMZ, public facing servers

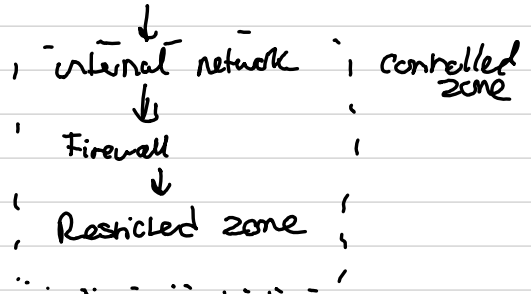
↳ internal network, private servers

↳ Restricted zone

uncontrolled zone

! internet → Firewall → DMZ → Firewall

! : : : : !



Classless inter-domain routing (CIDR) is a method of assigning multiples to IP addresses to create a subnet.

CIDR IP addresses are formatted like IPv4 addresses with a / and a number at the end.

Proxy Servers fulfil client requests by forwarding them to another server. Proxy Servers use temporary memory for commonly requested resources.

Forward Proxy - regulates and restricts access to the internet.

Reverse Proxy - restricts internet access to internal servers.

Module 3

Denial of Service attacks

disrupt a network by flooding it with traffic to overwhelm the network or prevent legitimate users being able to access it.

a DDoS attack uses multiple devices to target the network.

SYN Flood attack - flood a network with handshake requests to block ports

ICMP Flood - repeatedly send ICMP packet to get responses.

Ping of death - oversized ICMP packet can crash a system

Packet sniffing

The practice of capturing and analysing packets going across the network.

Passive - packets are read in transit

active - packets are recorded and then forwarded.
They may be altered

IP spoofing

attacker changes the source IP of a system to pretend they are someone they are not.

On-path attack - attacker puts himself in between two devices to receive traffic

Reply attack - attacker delays or replies a packet at a later time

Spoof attack - combination of DDoS and spoofing.

Module 4

hardening is the process of reducing a system's attack surface to prevent intrusion or compromise

Hardening is conducted on:

- OSes
- applications
- hardware
- servers
- databases

Pen-test - Simulated attack to find vulnerabilities

Configuration checking - accessing system configurations to find potential vulnerabilities

OS hardening

The OS is the interface between computer hardware and the user.

Patch update - a update that aims to resolve vulnerabilities

Something you have
Something you knew
Something you are

Network hardening

- Port filtering
- Privileges
- encryption
- Path codes
- rule analysis
- log analysis

You should use a SIEM.

Networks should be segmented

Cloud hardening

Consider IAM, configurations and the current attack surface

Note Shared responsibility

hypervisors come in two types:

type 1 - on the hardware

type 2 - on the software